

Exhibit 4-1.

Assessment Method #1 (Direct / Formative):

Rubric-based assessment of a major assignment in required courses.

Sample Description: Spring 2020 Assessment assignment, rubric and results for CHE 301.

ChE 301 – Chemical Engineering Thermodynamics –Spring 2020**Student Learning Outcome 1**

Rubric for assessment problem on final exam (Problems #1, 2, 4, 5)

- 1. Tank Pressurization Calculations with Redlich-Kwong Equation of State.**
- 2. Air Dew Point Calculations**
- VLE Data Reduction with the Peng-Robinson Equation of State
- 4. Excess Enthalpy Models for Isopropanol-Water Mixtures**
- 5. VLE Data Reduction with the NRTL Model**

SLO 1 Performance Indicator (PI)	(1) Unsatisfactory	(2) Developing	(3) Satisfactory
1. Apply mathematical and scientific principles to formulate models and systems relevant to the subject Problem 1A,1B 100 pts	Does not understand the connection between mathematical models and the system or process to be analyzed or designed (<50 pts)	Chooses a mathematical model or scientific principle that applies to an engineering problem, but has trouble in model development (50-70 pts)	Able to successfully combine mathematical and/or scientific principles to formulate models and systems relevant to chemical engineering (>70 pts)
2. Solve engineering problems by using the concepts of algebra, calculus, differential equations and/or linear algebra Problem 4A,4B,4C 100 pts	Does not understand the application of algebra, calculus, differential equations and/or linear algebra in solving chemical engineering problems (<50 pts)	Shows nearly complete understanding of applications of calculus and/or linear algebra in problem-solving (50-70 pts)	Applies concepts of algebra, calculus, differential equations and/or linear algebra to solve chemical engineering problems (>70 pts)
4. Translates academic theory into engineering applications Problem 2,5A,5B,5C 100 pts	Does not appear to grasp the connection between theory and the problem (<50 pts)	Some gaps in understanding the application of theory to the problem and expects theory to predict reality (50-70 pts)	Translates academic theory into engineering applications and accepts limitations of mathematical models of physical reality (>70 pts)

ChE 301 – Chemical Engineering Thermodynamics –Spring 2020

Student Learning Outcome 1

RESULTS BY PERFORMANCE INDICATOR			
Rubric #	1	2	4
Mapped SO	1	1	1
Average	2.56	2.78	2.44
Std Dev	0.78	0.65	0.86

Average scores were fairly uniform although there was some variation among students. More students were strong in solving engineering problems using theoretical thermodynamic concepts, and concepts of algebra, calculus, differential equations and/or linear algebra. Overall performance was satisfactory for all performance indicators.

Problem 1: Tank Pressurization Calculations with the Redlich-Kwong EOS

A cylindrical tank with volume $\{V = 25\text{ m}^3\}$ is originally filled with nitrogen at conditions of $\{T_o = 25^\circ\text{C}$ and $P_o = 1\text{ bar}\}$. The tank must be pressurized to a pressure of $\{P = 25\text{ bar}\}$ using a high pressure line which supplies nitrogen at a constant temperature and pressure equal to $\{T = 25^\circ\text{C}$ and $P = 50\text{ bar}\}$.

- A. Prove that by combining the dynamic energy and mass balances we obtain the equation:

$d(MU) = H dM$ where M is the mass(in moles) of nitrogen inside the tank at any time, U is the molar internal energy of nitrogen inside the tank and H is the molar enthalpy of the nitrogen in the high pressure line feeding the tank

- B. Integrate the equation obtained above and calculate the amount of nitrogen added to the tank during the pressurization process using (a) the ideal gas law and (b) the Redlich-Kwong equation of state. What is the error in assuming that the gas is ideal?